

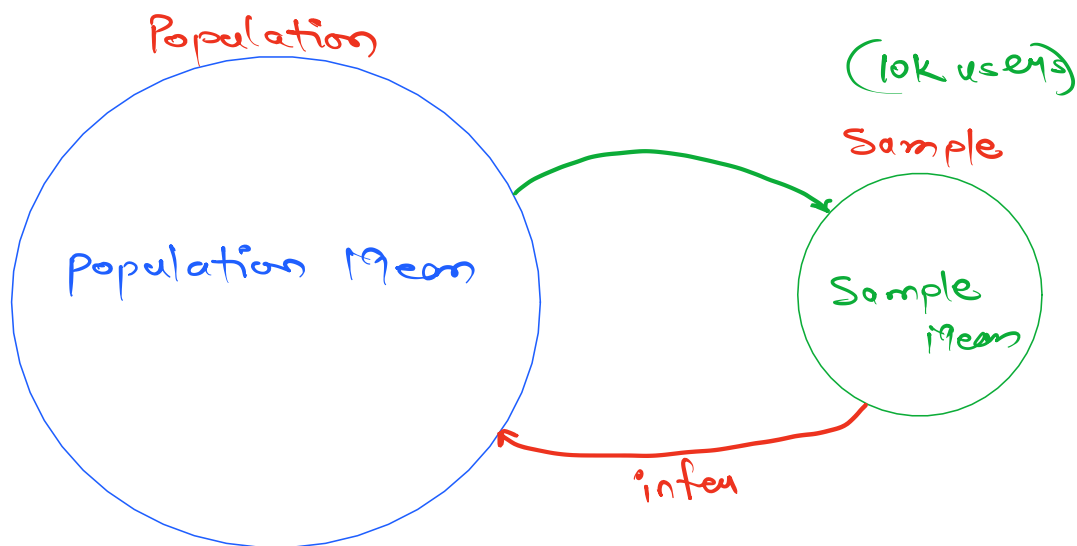
There are 2 branches in statistics:

① Descriptive statistics

② Inferential Statistics ←

- Population & Sample
- Confidence Interval
- Hypothesis Testing

Inferential statistics



Youtube: ^{daily} Avg. watchtime of users:

Population: All users of Youtube.
(2 billion)
(Pop size)

Population Data: Daily watchtime of every user in the population

- ① 83 mins
- ② 97 mins
- ③ 191 mins
- ⋮
- (2B) 200 mins

↓
Pop Mean = 195 mins/day

→ Time Constraint

→ Budget Constraint

What are the ways to estimate population mean using Sample Mean?

① Point Estimation

Population Mean = Sample Mean

Point Estimate

② Confidence Interval:

Confidence Interval = Point Estimate $\pm$ Margin of Error

S_1
 $\downarrow$
185 mins/day

No

= Sample Mean $\pm$ MOE

= 185 mins/day $\pm$ 5 mins/day

CI = 180 mins/day to 190 mins/day
interval

Avg Daily watchtime of users
is b/w 180 mins/day to 190 mins/day.

Prob:

Seagate/HDD/
Sandisk

HDD/SSD

How much should be the optional warranty
given to the customers?

Pop: All HDD manufactured by them.
(200K)

—
Why it is called confidence Interval not just

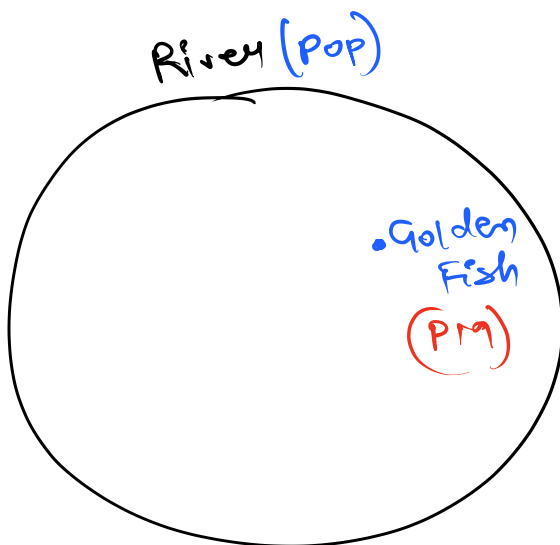
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Confidence Level

90%

95%

99%



Interval :

Interval

185 mins $\pm$ 5 mins

185 mins $\pm$ 7 mins

185 mins $\pm$ 10 mins

Hook ~~X~~

Fishing Net (CI)

Small

Med

Large

100%

20% — 90%